

A Novel Piezo-Based Technology for Haptic Feedback for XR

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ABSTRACT

We present a novel technology for tactile haptic feedback tailored to Virtual Reality (VR) and Augmented Reality (AR) experiences. The aim of this work was to develop and evaluate a novel technology which we first evaluate for its applicability for tactile haptic feedback (THF); in a later step we aim to build a glove-based system combining tactile and kinesthetic haptic feedback (KHF) based on the same technology. Unlike piezoelectric benders, eccentric rotating masses (ERM), and linear resonant actuators (LRA) we present an application of piezoelectric motors aimed at providing realistic tactile haptic feedback. In this research demo, we explain the context of application, the principle of operation, specifics to the implementation, and a technology evaluation. In the last section, we will sketch possible next iterations for an even more immersive haptic experience.

Index Terms: Hardware—Hardware validation—Physical verification—Design rule checking; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices;

1 INTRODUCTION

The promises of Virtual Reality (VR) and Augmented Reality (AR) are numerous, from increased productivity in virtual work environments [6], [1], to improved retention of muscle-memory due to higher engagement compared to traditional learning methodologies [3]. Immersion in these virtual environments is composed of multiple factors [2]: Display fidelity, motion tracking fidelity, and the visualization of a user's avatar. To further enhance the feeling of immersion, users should not only be able to see their virtual hand accurately tracking their actual hand motion, but they must also receive haptic feedback of the virtual hand touching a virtual object [7].

There are multiple possibilities to provide this feedback to a user immersed in VR, such as tactile, kinesthetic, thermal, and olfactory feedback. Of these, the most prominent are tactile haptic feedback (THF) and kinesthetic haptic feedback (KHF) [4]. KHF usually restricts a user's concentric finger flexion, either actively or passively [5]. THF, on the other hand, aims at simulating tactile sensations of users touching virtual surfaces. To achieve this, technologies such as eccentric rotating masses (ERM) and linear resonant actuators (LRA) are usually used, which are capable of vibrating at various amplitudes and frequencies to simulate various interactions (see, e.g., the home button of an earlier version of the iPhone). A unique technology is presented by HaptX' devices: A grid of pneumatically actuated air pockets touch users' fingertips and stimulate multiple points of interaction simultaneously.

In this research demo we focus on THF: We present a novel technology to create tactile haptic sensations at users' fingertips. Unlike vibrating masses, our technology is capable of applying a

force orthogonal to users' fingertips. Furthermore, the mechanical footprint of our technology allows us to place it both on the fingertip and on the most distal point on the finger that is in axis (the part with which one typically pushes keyboard keys or buttons).

Improving haptic fidelity is essential for professional training applications, as it is crucial for creating an immersive experience. Without it, users may feel disconnected and unable to properly understand their surroundings. Low-fidelity haptic feedback may be sufficient for entertainment applications such as games, but training real-world tasks in a virtual setting requires users to be able to interact with their environment at a high level of accuracy.

In the real world, a significant portion of human motion involves self-occlusions: For instance, when pointing or grabbing, users rely on their proprioception and haptic feedback to determine where their fingers are and how to interact with their surroundings. Relying on visual feedback alone can be unnatural and limit the impact of a virtual training scenario.

Haptic technology such as vibrating masses can provide discrete feedback to the user, i.e., whether or not they are touching a button, but they cannot create a force orthogonal to the fingertip. ERM and LRA technologies can provide users with the feeling of pressing against a surface, but they cannot be changed dynamically. This is problematic as real-world interaction is highly dynamic, and thus haptic fidelity needs to be improved in order to maximize the effectiveness of a virtual training simulation. Our research can contribute significantly to the fidelity of VR/AR simulations.

2 PRINCIPLE OF OPERATION

The underlying force and displacement generating unit is a piezoelectric motor manufactured by Miniswys. Piezoelectric motors are ideal for small-displacement applications that demand high precision, fast response times, and high speeds, such as those found in the optics of a smartphone camera system. By leveraging the power of piezoelectric motors, we were able to ensure highly accurate, low latency movements for the most precise and reliable operation.

Piezoelectric motors are typically driven with voltages between 3.3 V and 12 V and require minimal power. Depending on the mechanical design of the implementation, the motor unit can be optimized for either speed or force generation. For our application, we prioritize force generation, as the motor must eventually be able to push against the skin of a finger.

We have identified the rotation design as the optimal choice for our prototype due to its ability to prioritize force over speed. This design allows us to achieve greater precision and accuracy in our movements, ensuring that our prototype will meet all of our desired requirements.

3 IMPLEMENTATION

The mechanical design of the presented device is visualized in Fig. 1. In the center of the unit, a threaded cylinder is visible. Inside the casing two piezoelectric motors drive the counterpart of the threaded cylinder in either a counterclockwise or clockwise direction, causing the cylinder to move in or out of the unit.

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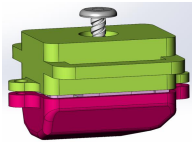
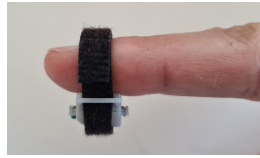


Figure 1: CAD design of our research prototype.



(a) Image of the haptic device.



(b) Device mounted on a user's finger. The very compact design is suitable for integration in gloves.

Figure 2: Images of the prototypes.

4 EVALUATION

- **Dimension:** The current implementation of the device has a base footprint of $8\text{ mm} \times 11\text{ mm}$ and a height of 8.2 mm .
- **Stroke:** The maximum stroke is 2 mm .
- **Speed:** The maximum speed is 15 mm/s .
- **Force:** The peak extrinsic force is approximately 1.3 N , the concentric force is approximately 0.55 N .

The height of the unit is an important factor for its use in VR gloves, as it can affect the pressure and force gradient when interacting with physical objects. To ensure the best fit for the mechanical casing to a fingertip and to find the optimum tradeoff between pressure/force gradient and casing size, we are currently evaluating the requirements on the maximum displacement of the unit. Depending on these requirements, smaller displacements might be found to be reasonable. For virtual object interactions, however, the height of the unit is not of primary concern, as users would almost never experience a situation where there is contact to a physical object. Only for specific gestures like closing hands or squeezing thumb and fingers, the height of the unit would play a noticeable role.

Speed has a significant impact on the time delay or latency of a haptic impulse being felt by users. It also affects the maximum frequency the unit can deliver to a finger surface. The current implementation's maximum speed is 15 mm/s , which requires 0.133 s from full retraction to full extension or vice versa, resulting in a maximum *full-amplitude* frequency of 3.75 Hz . To increase this frequency, the travel distance of the cylinder needs to be optimized. In the future, the casing design will be altered to accommodate this requirement.

The maximum force that the device can exert along the cylinder axis, combined with the area of the cylinder face that touches the finger, determines the maximum pressure that can be felt by users. In its current implementation, the tip of the cylinder has a radius of 1.5 mm , resulting in a surface area of 1.767 mm^2 . With the system currently capable of generating 1.3 N in-axis force, the maximum pressure felt is 106.7 lb/in^2 .

5 OUTLOOK

Our future work will depend on the outcomes of the current evaluation. We believe the baseline dimensions are satisfactory, as the device is small enough to be comfortably worn on even small fingers. One potential area of improvement is the height of the mechanical

unit, although its optimization will be contingent on the evaluation of speed and displacement amplitude.

One possible improvement to the current prototype would be to incorporate a multi-cylinder design. This would allow the unit to control multiple cylinders within a small area, allowing for greater efficiency and flexibility in the system. Similar to the grid-design for haptic feedback offered by HaptX, multiple cylinders could provide an even higher level of control and performance. Finally, further research into alternative build-ups for the unit could lead to even more options for customizing the design to suit different applications.

We will also conduct an evaluation of the haptic feedback to assess its effectiveness in improving user experience. We will compare the performance of the haptic feedback against competing technologies such as LRA, ERM, and HaptX to determine which provides the best user experience. Additionally, we will collect user feedback to gauge the users' perception of the haptic feedback. With this data, we can make further improvements to the haptic feedback system.

The ultimate goal will be to produce a glove which combines our established high-accuracy position and orientation tracking technology with effective haptic feedback, providing an invaluable instrument for several professional VR/AR applications.

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